RAMCO INSTITUTE OF TECHNOLOGY

2019-2020
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VOLUME 2
ISSUE 2

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

ELECOM MAGAZINSIO - TECHNICAL MAGAZINE

ABOUT THE DEPARTMENT

The Department of Electronics and Communication Engineering (ECE) is established in the year 2013 with a vision to Develop futuristic technologist in Electronics and Communication Engineering to meet the global challenges and the societal needs of our nation. At present RIT offers a 4-year B.E. ECE programme with a sanctioned intake of 120 students. The Laboratories are equipped with latest technology in the fields of Electronic Devices and Circuits, Digital Electronics, Linear Integrated Circuits, Microprocessors & Microcontrollers, Very Large Scale Integration Circuits, Digital Signal Processing, Communication System, Computer Networks, Optical & Microwave, Embedded Systems that offer opportunities on a wide range of hardware and advanced software packages. In addition to this, the Department has equipped with Research Laboratories such as NI LabVIEW Research Laboratory, Tessolve Semiconductor Laboratory and Technical Incubation Centre to enrich the knowledge of students and faculty.

The Department encompasses professional associations such as Students Technical Association "Electronica", IETE, ISTE chapters which provides a platform to learn professional and soft skills. The Department had organized various Value Added Courses, Electronica-Association activities, Workshops, Conference, Guest lectures and Industrial Visits. The students had excelled in both co-curricular and extra curricular activities. The faculty members also contributed their efforts to get sponsorship from FAER, TNSCST etc. The Department delights to release the Technical Magazine - "ELECOM MAGAZINSIO" for the academic year 2019-2020 Even Semester.

RAMCO INSTITUTE OF TECHNOLOGY

VISION

To evolve as an Institute of international repute in offering high-quality technical education, Research and extension programmes in order to create knowledgeable, professionally competent and skilled Engineers and Technologists capable of working in multi-disciplinary environment to cater to the societal needs.

MISSION

To accomplish its unique vision, the Institute has a far-reaching mission that aims:

- To offer higher education in Engineering and Technology with highest level of quality, professionalism and ethical standards
- To equip the students with up-to-date knowledge in cutting-edge technologies, wisdom, creativity and passion for innovation, and life-long learning skills
- To constantly motivate and involve the students and faculty members in the education process for continuously improving their performance to achieve excellence

QUALITY POLICY

RIT aims to impart Quality Education in Engineering and Technology through an effective teaching learning process, up-gradation of facilities and human resources, collaborating with Industry for promoting training and placement, research and consultancy activities with a commitment to continual improvement of Quality Management System.

ELECTRONICS AND COMMUNICATION ENGINEERING

VISION

Develop futuristic technologist in Electronics and Communication Engineering to meet the global challenges and the societal needs of our nation.

MISSION

- Educate students to become enlightened engineers with the latest know-how to take on the challenges concerned with the society.
- Equip the students with the current trends and latest technologies in the field of Electronics & Communication Engineering to make them technically strong and ethically sound.
- Constantly motivate the students and faculty members to achieve excellence in Electronics & Communication Engineering and Research & Development activities.

PPROGRAM EDUCATIONAL OBJECTIVES (PEOs) :

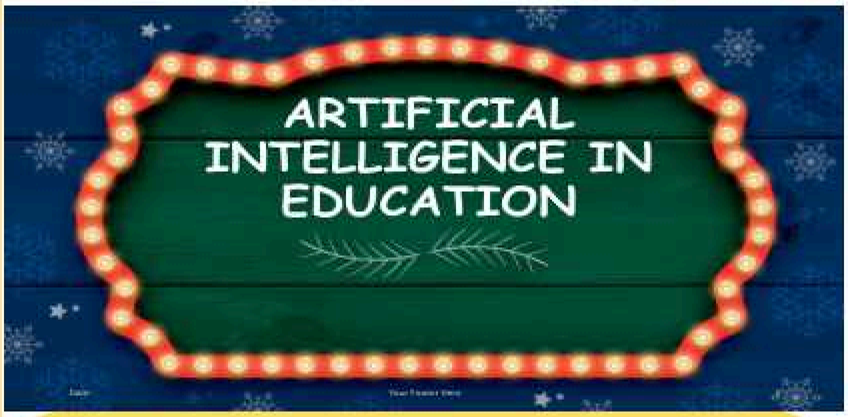
- Graduates will gain knowledge through continuous learning to enhance competency and career prospects.
- Graduates will be technically competent to design, analyze, develop and implement Electronic and Communication Systems.
- Graduates will apply the basic fundamental concepts in Electronic and related fields to solve general Engineering problems related to societal needs.
- Graduates will be professionals with attributes, passion, positive attitude, ethics and moral values for a successful career.
- Graduates will Communicate effectively and interact responsibly with colleagues, employees, team members, clients and the society.

PROGRAM SPECIFIC OUTCOMES (PSOs):

After successful completion of the degree, the students will be able to:

- Adapt to emerging trends in Information and Communication Technology by innovating new ideas to solve existing/novel problems.
- Excel in latest technologies of Embedded Systems.
- Design, analyze and develop electronic products in the area of VLSI Design and Communication Systems.
- Contribute as Entrepreneurs in building electronic products.

INSIDE THIS ISSUE



Introduction:

Artificial intelligence (AI) is the imitation of human intelligence processes such as speech and visual recognition, translation of the languages and virtual decision-making by machines and robots. The ability of machine to think and behave like human beings, has given AI a special place in all fields. AI is present everywhere in various aspects of our lives starting from intelligent sensors to personal assistants. Recent advancements in AI have brought in many tremendous changes in the educational field. Artificial intelligence helps students and teachers to make their educational experience wonderful.

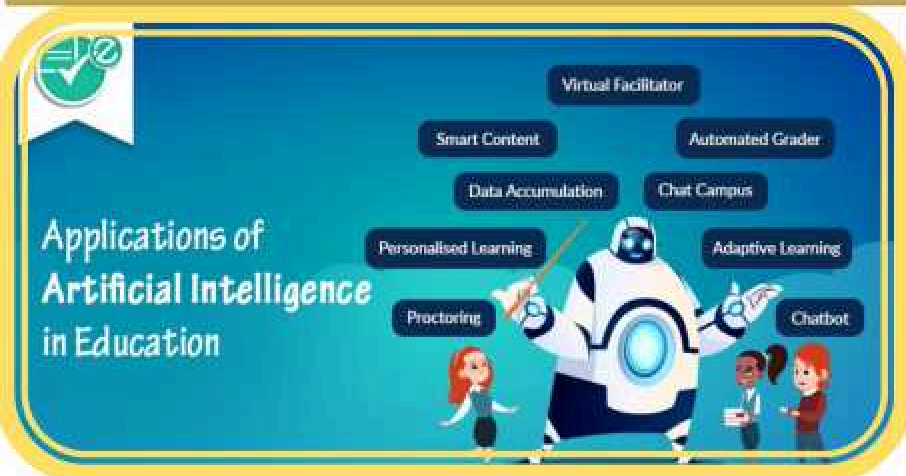
BY

Ms.S.Mahalakshmi,
III Year ECE
Ms.R.Leon Mobit Bringa,
III Year ECE

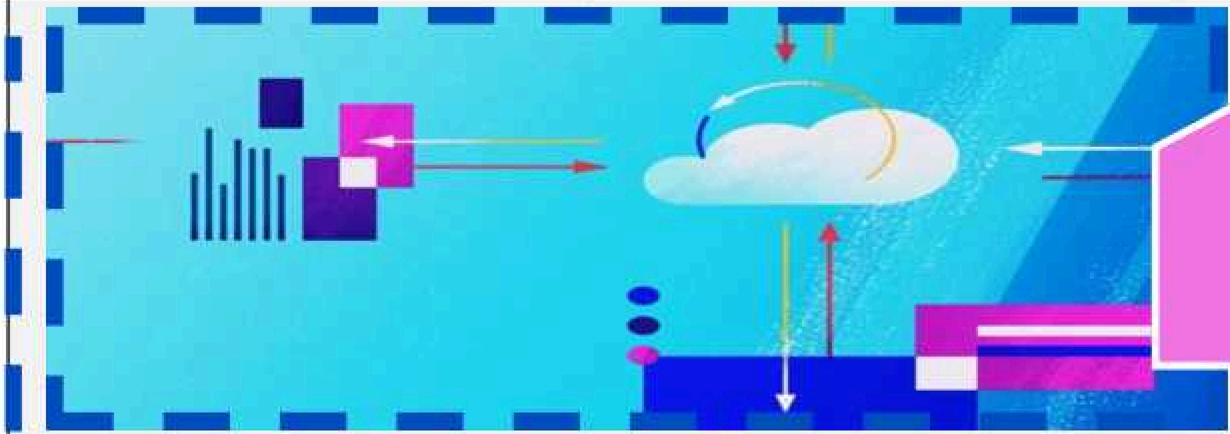
Source: https://pub.towardsai.net/artificial-intelligence-in-education-benefits-challenges-and-use-cases-

Now let's have a look on the role that it plays in Education:-

- AI is used in grading system where teachers can automate grading for certain type of questions.
- It will be more useful in adaptive and individualized learning which satisfies the needs of the students.
- AI helps the teachers to know the understanding capacity of the students on their lectures and enable them to provide the relevant hints for students.
- It acts as a tutor for the students and helps them to learn. Artificial intelligence driven programs offers helpful feedback for both students and teachers.
- It helps the teachers to monitor the performance of the students and enable them improve the instruction that they provide for the students.
- AI systems in schools have changed the way students find and interact with integrated technology.
- This has an effect to change teachers as facilitators by providing students interactive learning and provide assistant for their improvement.



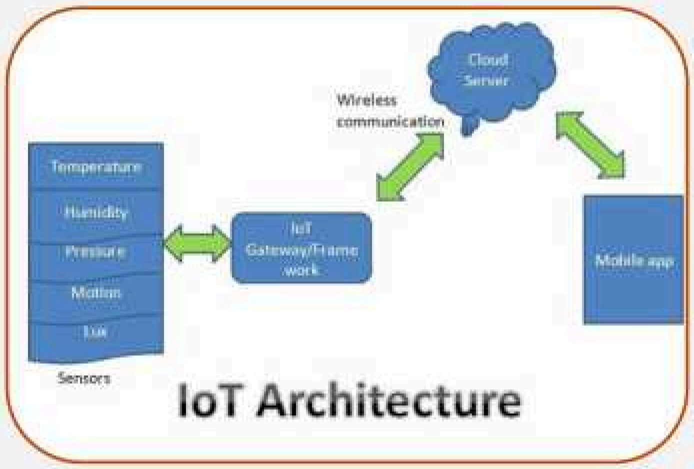
- Artificial intelligence systems acquired data will change the way the schools find, teach and support students.
- If goes beyond the reach it may even re-place teachers in certain instance. It has become a learning companion the assists students in their learning process.



INTERNET OF THINGS (IoT)

The Internet of Things (IoT) is the network of physical objects or things embedded with electronics, software, sensor and network connectivity which enables these objects to collect and exchange data.

Working of IoT:



The Internet of Things (IoT) also sometimes referred to as the internet of everything, consist of all web-enabled devices that collect, send and act on data they acquire from their surroundings environment using embedded sensors, processors and communication hardware. These devices, often called "connected" or "smart" devices , can sometimes talk to other other related devices, a process called machine to machine communication, and act on the information they get from one another. Humans can interact with the gadget to set them up, give them instruction or access the data, but the devices compost of the work on their own without human intervention. Their existence has been made possible by all the tiny mobile component that are available these days, as well as the always online nature of our home and business networks.

A things in the context of the internet of things is an entity or

physical object that has a unique identifier, an embedded system and ability to transfer data over a network. For Example:

- Heart monitoring implants.
- Biochip transponders on farm animals
- Automobiles with built-in sensors
- DNA analysis devices & other wearable's etc.

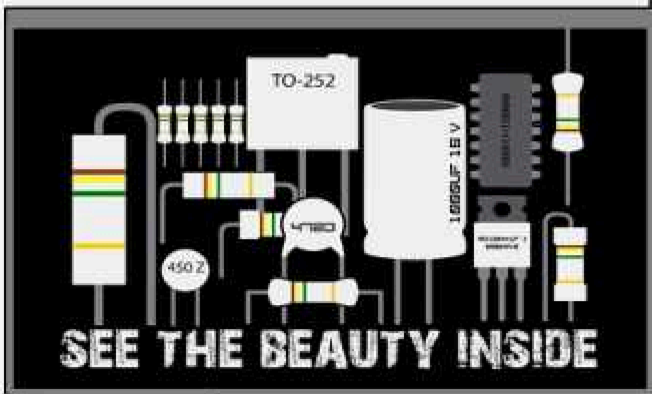
Application of IoT

These devices collect useful data with the help of various existing technologies and then autonomously flow the data between other devices.

By

Ms.S.Uvasri, II Year ECE

Source: **https://www.guru99.com/iot-tutorial.html#3**



INDUSTRY 4.0 DURING COVID-19

By

Mr.M.Sathis Kannan, II Year ECE

In the past few decades, a fourth industrial revolution has emerged, known as Industry 4.0. Industry 4.0 takes the emphasis on digital technology from recent decades to a whole new level with the help of interconnectivity through the Internet of Things (IOT), access to real-time data, and the introduction of cyber-physical systems. Advanced Industry 4.0 technology to help its global operations safely navigate the ongoing COVID-19 crisis and continue to service its customers. Industry 4.0 empowers business owners to better control and understand every aspect of their operation, and allows them to monitor instant data to boost productivity, improve processes, and drive growth.

As during this pandemic many company's changes their production through augmented reality and they able to continue to support operation remotely and continue the development of product aspect of customer deadline to met despite the global challenges we are all facing.

Here how the many company's using industries 4.0 on this pandemics:

1. Displaying 3D images and connecting remotely to improve safety

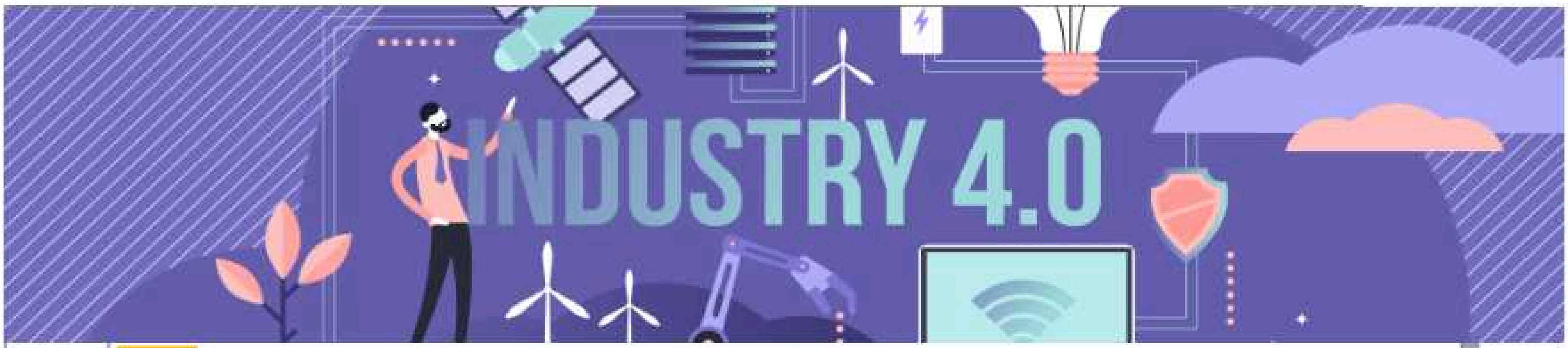
- To enable remote assistance and critical activities, company's are using Microsoft's HoloLens 2 augmented reality goggles that offer the capability to display 3D images in physical spaces and connect remotely. This remote assistance technology ensures that ongoing work can be performed while keeping everyone safe.

2. Enhancing training and expediting review processes through augmented reality

- Augmented reality also assists with knowledge retention and ongoing training. For instance, the company's identified an opportunity to use the technology to train operators and engineers on new equipment, which would have traditionally been conducted by instructors who travel to the various remote sites.

3. Implementing robots to promote safe distancing and increase productivity

- To optimize manufacturing flow and eliminate the need for forklifts and other human-operated transport machinery, the company's is using autonomous automated guided vehicles (AGVs) or autonomous mobile robots (AMRs) which increase safety and allow for social distancing while also lowering costs.



4. Leveraging 3D printing tools internally to expedite processes

- Additive manufacturing is leveraged to improve safety, quality and efficiency by designing and producing tools, poke-yokes and gauges internally on both polymer and metal materials. This allows fast reactions (shorter lead times), reduced purchase costs and highly customized solutions.

Following are the Industry 4.0 need to overcome from this issues:

a. The IoT security is a major concern. The companies are working to address the security related loop holes.

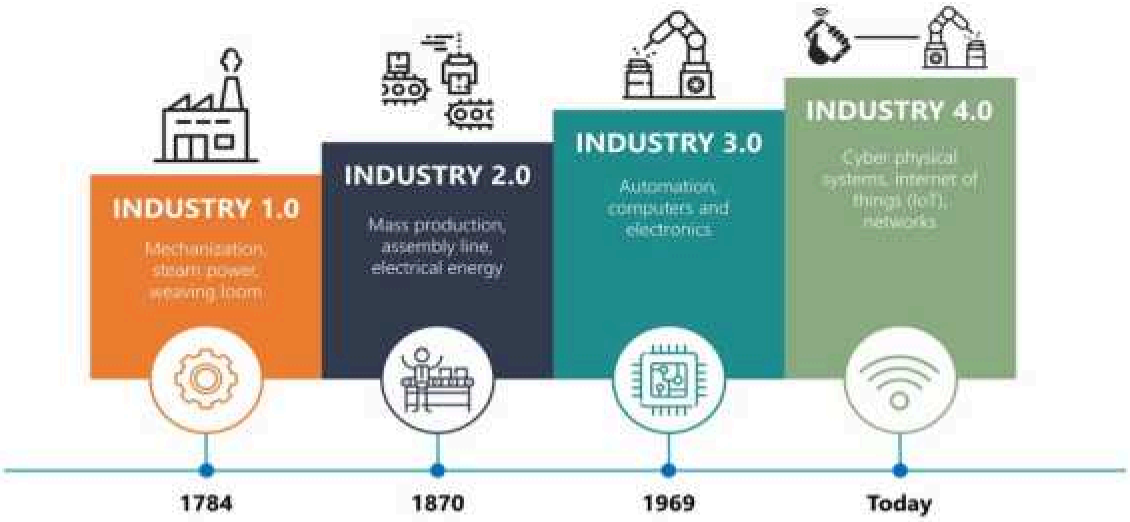
b. Skills and education of workers working on industry 4.0 based processes need to improved.

c. It requires to maintain integrity of production processes.

d. Reliability and stability are required for M2M (Machine to Machine) communication.

As industries 4.0 change this pandemic period into more advancement of technology on real world. Industries 4.0 allows you to address potential issues before they become big problems and allows you to trim costs, boost profits, and fuel growth. Industry 4.0 technology helps you manage and optimize all aspects of your manufacturing processes and supply chain. But still it not sure that Industries 4.0 is right for us.

The Industrial Revolution: From Industry 1.0 to Industry 4.0:



Source: https://www.emeraldgrouppublishing.com/archived/products/journals/call_for_papers.htm%3Fid%

BY
Ms.V.Alphina Christy
Ms.S.Krishna Veni
II Year ECE

EMERGING COMPANIES ON A MISSION

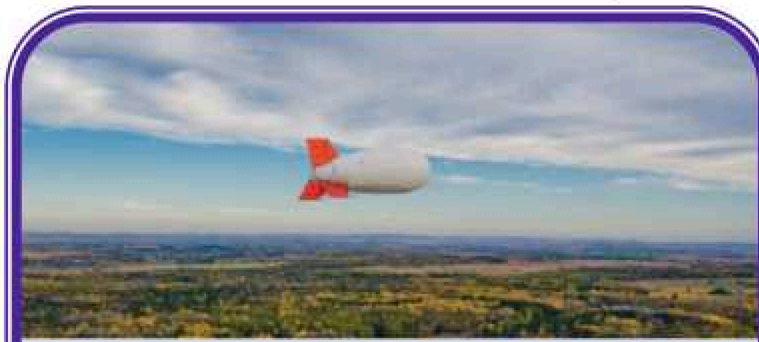
Biology-guided radiotherapy, crop-protecting drones, broadband super towers, intelligent electric scooter are combining novel technologies with bold vision. Engineers at emerging companies are on a mission to improve the lives of individuals and entire communities.

1. Riding the Wave of Industry 4.0

Industrial plants are gathering increasing amounts of data as they move toward full interconnectivity and automation. Until now, this data could only be accessed on a desktop or tablet. GlassUpF4 augmented reality (AR) visors enable machine operators to access the data on the shop floor and without stopping work. The visors include a video camera, an LED lighting system, voice control, and Bluetooth, as well as a gyroscope, compass, and accelerometer. A remote control dashboard allows the user to share their point of view with any paired visor for on-the-job training and remote maintenance.

2. Beaming Mobile Broadband to Isolated Communities

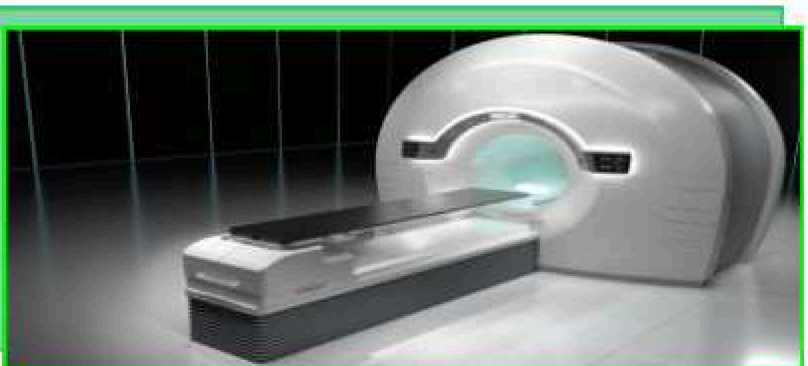
Altaeros plans to turn tethered blimps, or aerostats, into autonomous telecommunication SuperTowers that deliver high-speed broadband to the 4 billion people worldwide without access to the internet. One SuperTower could service an area that would require up to 20 conventional cell phone towers. Altaeros has completed pilot tests and is working with the FCC and the FAA to meet all US federal requirements to operate the Super Tower commercially.



"Our hope is that by accelerating the rollout of rural telecom infrastructure, we can help bring underserved communities online for the first time."
– Igor Braverman, Altaeros

3. Turning Cancer on Itself

RefleXion Medical is developing the first-ever biology-guided radiotherapy system for cancer treatment. The system combines a PET scanner and a radiotherapy/radiosurgery machine in a novel way. By means of PET tracers, tumors continuously signal their location in real time during treatment. This capability could one day revolutionize cancer care by enabling clinicians to treat multiple tumors in a single session .



"Using biology to guide radiotherapy, we hope to have the means to turn cancer on itself."
– Sam Mazin, Ph.D., RefleXion

Source: https://in.mathworks.com/company/newsletters/articles/matlab-and-simulink-in-the-world-emerging-companies-on-a-mission.html

4. Solving the "Cocktail Party Problem"

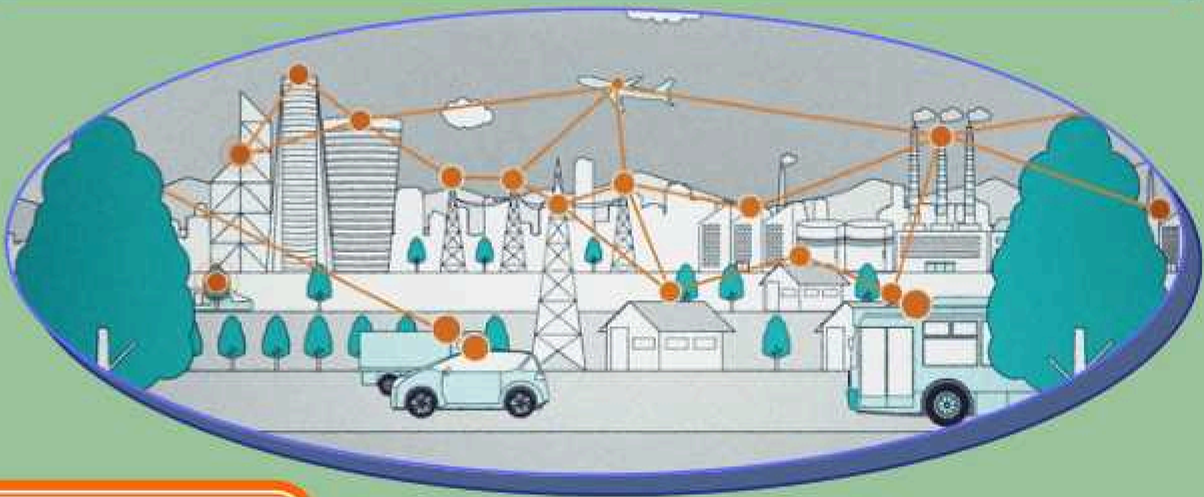
Using a hearing aid, smart home appliance, or other speech-recognition device in a noisy environment such as a party is challenging because the microphone has difficulty distinguishing one voice from all the others. Yobe's Voice Identification System for User Profile Retrieval (VISPR) combines artificial intelligence (AI) and signal processing to identify a signal's biometric characteristics, identify individual speakers, and distinguish speech from noise.

> *"The ways we have been interacting with machines up until now have been artificial because these machines haven't been able to hear us. The natural way to communicate with something is to talk to it."*
>
> – Ken Sutton, Yobe

5. Transforming the City Commute

Ather Energy's 450 electric scooter is the first of its kind in India. Designed for city driving, the Ather 450 can autonomously navigate tight parking spots, complicated street systems, and stop-and-go traffic. It is powered by a high-capacity li-ion battery pack, and has a charging range of 46 m and a top speed of 50 mph.

WIRELESS COMMUNIATION



By,
M.APINEYA SRI
A.BALA KIRUTHIKA
II Year ECE

Source: https://www.coursera.org/lecture/wireless-communications/1-2-first-to-second-generation-1g-to-2g-1us1e & https://old.amu.ac.in/emp/studym/100018546.pdf

Introduction :

The term wireless communication refers to the electromagnetic transfer of information between two or more points that are not connected by an electrical conductor.

- Transmitting/receiving data and voice using electromagnetic waves in open space
- The information from sender to receiver is carried over a well defined channel
- Each channel has a fixed frequency bandwidth and capacity (bit rate)
- Different channels can be used to transmit information in parallel and independently

Why we go for Wireless communication?

- Free from wires
- "Auto Magical" instantaneous communication without physical connection set up
- Global coverage
- Communication can reach were wire is infeasible or costly

Types of Wireless communication

The different types of wireless communication technologies include

- IR Wireless communication
- Satellite communication
- Broadcast radio
- Microwave radio
- Bluetooth
- Zigbee etc.

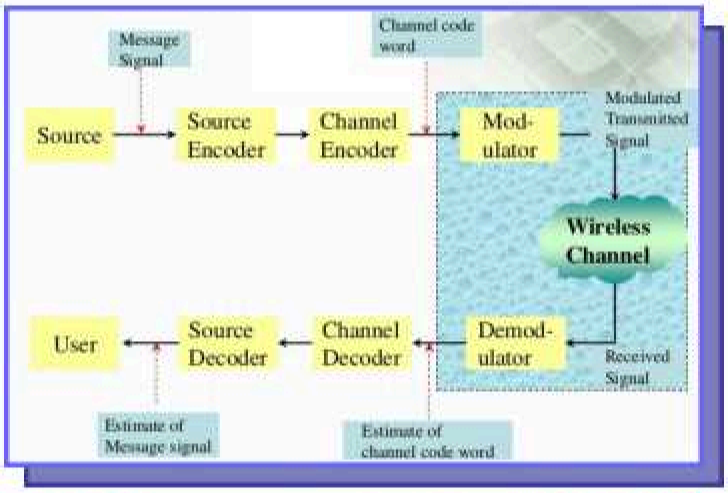
WIRELESS COMMUNIATION

Basic elements of wireless communication systems:

A typical wireless communication system can be divided into three elements. They are,

- Transmitter
- The channel
- Receiver

Block Diagram:



Cellular communication (1G to 4G LTE)

1G was introduced in 1980's . The key technologies of 1G is cellular structure, analog FM and FDMA. The most important development in mobile communication history, is the cellular communication 1G was designed to send voice messages only on the other hand we cannot send SMS, video call etc. To overcome this, 2G was introduced in 1990's.

The key word of 2G is digital transmission. Here, Voice communication quality was dramatically improved when compared to 1G and people were able to transmit data rather than just making a voice call. This 2G system using BPSK and QPSK as modulation technology. By using BPSK and QPSK ,we can send one bit and two bits per symbol respectively .In satellite and military communication, they use 2G system. The data rate of this is 264kgs.However, the data rate of 2G is still limited and so 3G came into existence .

3G was introduced in 2000's. Key word of 3G is only data and data transmission. The data rate was increased upto 10 mega bps compared to 2G .But the data rate is still limited to real time high quality video conferencing. So the effort begun to develop a new system. That system is 4G LTE.

4G was first developed in 2010. It uses much BW than 3G.in 4G we are using more than 20MHz upto 100MHz. OFDMA is the most important development in LTE. Here SMS, Voice call, Video call is possible

History of Cellular Systems

Generation	Time	Technology	Features
0G	1947	Mobile Telephone Service (MTS) by AT&T	Analog Heavy (36 kg) ~5000 customers
1G	1983 (NTT had an earlier deployment in 1979)	Advanced Mobile Phone System (AMPS)	Analog FDMA Voice
2G	1991	Global System for Mobile Communication (GSM) in Europe (Finland)	Digital FDMA Voice
3G	2001	WCDMA CDMA2000 1xEV-DO HSDPA	Digital CDMA Broadband Data
4G	2010	WiMax LTE	Digital OFDMA Data, IP network

WIRELESS COMMUNIATION

Why we need 5G?

We all are familiar about Internet of things, virtual reality, 3D video communication and 3D games and so on. The 4G systems cannot handle those services.SO We need a power system, that's 5G, which is 20 times faster than 4G and the connectivity increases by 10 times than 4G and the speed is 500 kilometre per hour.

Applications of Wireless Communication:

Applications of wireless communication involve security systems, television remote control, Wi-Fi, Cell phones, wireless power transfer, computer interface devices and various wireless communication based projects.

Advantages of wireless communication:

- Any data or information can be transmitted faster with a high speed
- Maintenance and installation is less cost for these networks.
- The internet can be accessed from anywhere wirelessly
- It is very helpful for workers, doctors working in remote areas as they can be in touch with medical centres.

Disadvantages of wireless communication:

- An unauthorized person can easily capture the wireless signals which spread through the air.
- It is very important to secure the wireless network so that the information cannot be misused by unauthorized users

R.Anusuya, II Year ECE

All India Essay Contest – April 2020
National Development Agenda

HILL GROVE
RESEARCH PVT LTD

An Essay on Social Issues

Submitted by R.Priyadharshini, III Year ECE

INTRODUCTION:

India is a famous country all over the world. It is so famous for its culture and unity. India is a democratic country. I am very much proud to say "I am an Indian". Now, for every situation or any matter, ideas and career itself while taken into account, there must be some irrelevant things may happen. Likewise, India has some social issues. Even god can made a mistake right!!!. Everyone is not perfect in everything. These social issues and orders taken by the government but also efforts must be taken by each and every individual.

PRESENT SCENARIO:

"Alone we can do little, together we can do so much" - Helan

As we are in developing country, the present India reveals so many efforts. Out of 100, 40% of people understand their individualism towards India. But, Remaining are not have an awareness towards society, its due to uneducation. But, if educated people also do the same, then what to tell!!!. See, for example, the current scenario: COVID-19 revolves his world, but every individual should cooperate with law to fight against that virtual ghost. But, some are not follow the rules. They just are roaming here and there without any responsibility. This is our country. All are brothers and sisters. The Problems get afraid and panic. It will happen by our own involvement towards society. So, everyone should do this role correctly to be the best of our society.

KEY ISSUES AND CHALLENGES:

"The strength of the team is each individual member.

The strength of each member is the team"

As we all know, children follow the habit of elder who revolves around them. This means that they observe their parents, elders and their teaching towards life. It reflects them to behave in this society. Likewise, if they preach the God, they follow the words of God. Hence, each and every officers should treat people, citizen as children and they should be the role model towards society. See, if every officers do their work sincerely, then no issues will be there. Officers in the sense, everyone, Prime minister, Chief Minister, MLAs, Collector, Police, Doctor, Engineers, Teachers, Salesman, Student etc. Peoples follow them to lead our society. If they do their work correctly, people automatically strict to their rules. Every individual should start their work towards development.

SOLUTIONS AND SUGGESTIONS

"Every problem must have a solution"

I love Maths. It teaches me this wonderful lesson. Continuous motivation towards "Individualism is important" for the growth of country. Everyone should involve towards it. Small changes itself brings good result. If any plastics in the roadside throw it in the appropriate dustbin. Small changes leads to big changes. So, motivational speech should be given to every city in the country and also rewards should be given for the best "Society lover" . Employed and educated students should advise the uneducated towards the improvement of society .

In a home, atleast one educated person will be there. If he/she tries to understand and preaches his family, it becomes big challenge leads to revolution in our society. Starting point may come from anywhere, but the result is crucial. Make public places clean and green.

CONCLUSION:

"Individual commitment to a group effort. That is what makes a team work, a company work, a society work and civilization work."

As we are student and youth, we are the most valuable asset for our country. We should take our own step to lead our country in this world. If any illegal or issues, won things happen, we should come forward to inform it the officials without any panic. Be bold. "Proud to be an Indian". Let say "India is youth country" .

Sl. No.: 20E0143

All India Essay Writing Contest – April 2020

CERTIFICATE OF PARTICIPATION

Presented to

Priyadharshini R, BE (ECE)

Ramco Institute of Technology, Rajapalayam, Tamil Nadu

for participating in the National Level Essay Writing Contest for Collegiate Students held on 30th April 2020.

HILLGROVE RESEARCH PVT LTD
Coimbatore, Tamil Nadu

05/05/2020

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